

GraphTheorySymbol: $\text{\LaTeX}$ Commands with Symbols for Graph Theory

Laurent Frédéric Bernard François Lyaudet

Website: <https://lyaudet.eu/laurent/>

Email: Laurent.Lyaudet@gmail.com

v1.7.0 2026/09/21

Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

Contents

1	Introduction	3
2	Usage	4
2.1	Memo or for the impatient	4
2.2	Detailed usage	7
2.2.1	General macros	7
2.2.2	Standard adjacency macros	7
2.2.3	Standard incidence macros	11
2.2.4	Monochrome adjacency macros	14
2.2.5	Monochrome incidence macros	18
2.2.6	Meta-logic macros	22
2.3	Good practices	23
3	Implementation	24
4	Acknowledgments	37
5	FAQ	37
6	Support	37
7	Change History	38

1 Introduction

This documentation is for $\text{\LaTeX}$ package `GraphTheorySymbol`. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com/LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$

Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSOIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

<code>\GTSsetMainColor{blue}</code>			
Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>		$v \overset{\circ}{\sim} e$

Long name	Short name	Result	In context
<code>\GTSmetaNot/</code>	<code>\GTSmN/</code>	$\neg$	$\neg \overset{\circ}{\sim}$
<code>\GTSmetaAnd/</code>	<code>\GTSmA/</code>	$\wedge$	$\overset{\circ}{\sim} \wedge \overset{\circ}{\sim}$
<code>\GTSmetaOr/</code>	<code>\GTSmO/</code>	$\vee$	$\overset{\circ}{\sim} \vee \overset{\circ}{\sim}$

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Directed graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Oriented graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Tournament adjacency:

$\overset{\circ}{\sim} \overset{\circ}{\sim}$ is obtained with `\mathrel{\{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/\}}`.

Tournament adjacency with maybe some colored edge also, who knows?:

$\overset{\circ}{\sim} \overset{\circ}{\sim}$ is obtained with `\mathrel{\{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/\}}`.

Directed graph:

$\overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim}$ is obtained with

`\mathrel{\{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/\GTSmO/\GTSdAD/\GTSdAU/\GTSmO/\GTSnDAD/\GTSnDAU/\}}`.

Oriented graph:

$\overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim}$ is obtained with

`\mathrel{\{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/\GTSmO/\GTSnDAD/\GTSnDAU/\}}`.

Prefix notation:

$\exists u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} v \Leftrightarrow \exists u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

$\exists v, e, v \overset{\circ}{\sim} \overset{\circ}{\sim} e \Leftrightarrow \exists v, e, v \overset{\circ}{\sim} \overset{\circ}{\sim} e$.

$\overset{\circ}{\sim} \overset{\circ}{\sim}$ is obtained with

`\mathrel{\{\GTSmO/\GTSxa[mC=red,d]\GTSxa[mC=green,d]\GTSxa[mC=blue,u]\}}`.

$\mathrel{\mathop{\vee}\limits^{\text{red,green,blue}}}$ is obtained with
`\mathrel{\{\GTSm0/\GTSxi[mC=red,i]\GTSxi[mC=green,i]\GTSxi[mC=blue,o]\}}.`

Apart from these symbol macros, versatile macros to call them all are available with options:
`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,
`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,
`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,
`\GTSx[]{} for all symbols`
 $(\GTSx[]\{a\} = \GTSxa[], \GTSx[]\{i\} = \GTSxi[], \GTSx[]\{m\} = \GTSxml[]).$
 See Detailed usage section for full syntax of these macros.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

For completeness, we also documented macros with an at sign, but as usual, you should not (have to) use them directly.

2.2.1 General macros

\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

2.2.2 Standard adjacency macros

\GTS@a This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@b**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

\GTS@b This macro builds on top of **\GTS@a** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

\GTSxadjacency, **\GTSxadj**, **\GTSxa** Versatile macros for adjacency that can be given options.

Argument:

`[\langle optionsList \rangle]` Sets the following options if not empty:

`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,

`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,

monochrome or mono or mon or m (equivalent to `mC=black,sC=black`),
 negated or not or n,
 up or u,
 down or d,
 relation or rel or r.

`\GTSadjacency`, `\GTSadj`, `\GTSa` This macro will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and `multimap` symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as `Adj(.,.)`.

`A.\GTSadjacency/A.`

or `A.\GTSadj/A.`

or `A.\GTSa/A.`

will yield $A \circ A.$

`u \GTSadjacency/ v`

or `u \GTSadj/ v`

or `u \GTSa/ v`

will yield $u \circ v.$

`$u \GTSadjacency/ v$`

or `$u \GTSadj/ v$`

or `$u \GTSa/ v$`

will yield $u \circ v.$

`\GTSadjacencyRelation`, `\GTSadjRel`, `\GTSaR` This macro will draw an adjacency symbol.

`$u\GTSadjacencyRelation/v$`

or `$u\GTSadjRel/v$`

or `$u\GTSaR/v$`

will yield $u \circ v.$

`\GTSunadjacency`, `\GTSunadj`, `\GTSu` This macro will draw an unadjacency symbol.

`A.\GTSunadjacency/A.`

or `A.\GTSunadj/A.`

or `A.\GTSu/A.`

will yield $A \not\circ A.$

`u \GTSunadjacency/ v`

or `u \GTSunadj/ v`

or `u \GTSu/ v`

will yield $u \not\circ v.$

`$u \GTSunadjacency/ v$`

or `$u \GTSunadj/ v$`

or `$u \GTSu/ v$`

will yield $u \not\circ v.$

`\GTSunadjacencyRelation`, `\GTSunadjRel`, `\GTSuR` This macro will draw an unadjacency symbol.

`$u\GTSunadjacencyRelation/v$`

or $\$u\backslash\text{GTSunadjRel}/v\$$

or $\$u\backslash\text{GTSuR}/v\$$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\backslash\text{GTSdirectedAdjacencyUp}$, $\backslash\text{GTSdirAdjUp}$, $\backslash\text{GTSdAU}$ This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUp}/A.$

or $A.\backslash\text{GTSdirAdjUp}/A.$

or $A.\backslash\text{GTSdAU}/A.$

will yield $A.\overset{?}{\underset{?}{\rightharpoonup}}A.$

$u \backslash\text{GTSdirectedAdjacencyUp}/ v$

or $u \backslash\text{GTSdirAdjUp}/ v$

or $u \backslash\text{GTSdAU}/ v$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\$u \backslash\text{GTSdirectedAdjacencyUp}/ v\$$

or $\$u \backslash\text{GTSdirAdjUp}/ v\$$

or $\$u \backslash\text{GTSdAU}/ v\$$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\backslash\text{GTSdirectedAdjacencyUpRelation}$, $\backslash\text{GTSdirAdjUpRel}$, $\backslash\text{GTSdAUR}$ This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyUpRelation}/v\$$

or $\$u\backslash\text{GTSdirAdjUpRel}/v\$$

or $\$u\backslash\text{GTSdAUR}/v\$$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyUp}$, $\backslash\text{GTSnegDirAdjUp}$, $\backslash\text{GTSnDAU}$ This macro will draw a negated directed adjacency upward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyUp}/A.$

or $A.\backslash\text{GTSnegDirAdjUp}/A.$

or $A.\backslash\text{GTSnDAU}/A.$

will yield $A.\overset{?}{\underset{?}{\rightharpoonup}}A.$

$u \backslash\text{GTSnegatedDirectedAdjacencyUp}/ v$

or $u \backslash\text{GTSnegDirAdjUp}/ v$

or $u \backslash\text{GTSnDAU}/ v$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\$u \backslash\text{GTSnegatedDirectedAdjacencyUp}/ v\$$

or $\$u \backslash\text{GTSnegDirAdjUp}/ v\$$

or $\$u \backslash\text{GTSnDAU}/ v\$$

will yield $u \overset{?}{\underset{?}{\rightharpoonup}} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyUpRelation}$, $\backslash\text{GTSnegDirAdjUpRel}$, $\backslash\text{GTSnDAUR}$ This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSnegatedDirectedAdjacencyUpRelation}/v\$$

or $\$u\backslash\text{GTSnegDirAdjUpRel}/v\$$

or $\$u\backslash\text{GTSnDAUR}/v\$$
 will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash\text{GTSdirectedAdjacencyDown}$, $\backslash\text{GTSdirAdjDown}$, $\backslash\text{GTSdAD}$ This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$\text{A}\backslash\text{GTSdirectedAdjacencyDown}/\text{A}$.

or $\text{A}\backslash\text{GTSdirAdjDown}/\text{A}$.

or $\text{A}\backslash\text{GTSnDAD}/\text{A}$.

will yield $\text{A}\overset{\curvearrowright}{\sim}\text{A}$.

$u \backslash\text{GTSdirectedAdjacencyDown}/ v$

or $u \backslash\text{GTSdirAdjDown}/ v$

or $u \backslash\text{GTSnDAD}/ v$

will yield $u \overset{\curvearrowright}{\sim} v$.

$\$u \backslash\text{GTSdirectedAdjacencyDown}/ v\$$

or $\$u \backslash\text{GTSdirAdjDown}/ v\$$

or $\$u \backslash\text{GTSnDAD}/ v\$$

will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash\text{GTSdirectedAdjacencyDownRelation}$, $\backslash\text{GTSdirAdjDownRel}$, $\backslash\text{GTSdADR}$ This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyDownRelation}/v\$$

or $\$u\backslash\text{GTSdirAdjDownRel}/v\$$

or $\$u\backslash\text{GTSdADR}/v\$$

will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyDown}$, $\backslash\text{GTSnegDirAdjDown}$, $\backslash\text{GTSnDAD}$ This macro will draw a negated directed adjacency downward symbol.

$\text{A}\backslash\text{GTSnegatedDirectedAdjacencyDown}/\text{A}$.

or $\text{A}\backslash\text{GTSnegDirAdjDown}/\text{A}$.

or $\text{A}\backslash\text{GTSnDAD}/\text{A}$.

will yield $\text{A}\overset{\curvearrowleft}{\sim}\text{A}$.

$u \backslash\text{GTSnegatedDirectedAdjacencyDown}/ v$

or $u \backslash\text{GTSnegDirAdjDown}/ v$

or $u \backslash\text{GTSnDAD}/ v$

will yield $u \overset{\curvearrowleft}{\sim} v$.

$\$u \backslash\text{GTSnegatedDirectedAdjacencyDown}/ v\$$

or $\$u \backslash\text{GTSnegDirAdjDown}/ v\$$

or $\$u \backslash\text{GTSnDAD}/ v\$$

will yield $u \overset{\curvearrowleft}{\sim} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}$, $\backslash\text{GTSnegDirAdjDownRel}$, $\backslash\text{GTSnDADR}$ This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}/v\$$

or $\$u\backslash\text{GTSnegDirAdjDownRel}/v\$$

or $\boxed{\$u\backslash\text{GTSnDADR}/v\$}$
 will yield $\boxed{u \nparallel v}$.

2.2.3 Standard incidence macros

\GTS@i This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

- $\{\langle mainColor \rangle\}$ If non-empty, this color will be used as main color.
- $\{\langle secondColor \rangle\}$ If non-empty, this color will be used as second color.
- $\{\langle drawSlash \rangle\}$ Non-empty implies that a negation will be drawn.
- $\{\langle drawUpArrow \rangle\}$ Non-empty implies that an up-arrow will be drawn.
- $\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.

\GTS@j This macro builds on top of **\GTS@i** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

- $\{\langle mainColor \rangle\}$ If non-empty, this color will be used as main color.
- $\{\langle secondColor \rangle\}$ If non-empty, this color will be used as second color.
- $\{\langle drawSlash \rangle\}$ Non-empty implies that a negation will be drawn.
- $\{\langle drawUpArrow \rangle\}$ Non-empty implies that an up-arrow will be drawn.
- $\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.
- $\{\langle relation \rangle\}$ Non-empty implies that relation spacing will be added.

\GTSxincidence, **\GTSxinc**, **\GTSxi** Versatile macros for incidence that can be given options.

Argument:

- $[\langle optionsList \rangle]$ Sets the following options if not empty:
 $mainColor=color1$ or $mainC=color1$ or $mColor=color1$ or $mC=color1$,
 $secondColor=color2$ or $secondC=color2$ or $sColor=color2$ or $sC=color2$,
 $monochrome$ or $mono$ or mon or m (equivalent to $mC=black, sC=black$),
 $negated$ or not or n ,
 in or i ,
 out or o ,
 $relation$ or rel or r .

\GTSincidence, **\GTSinc**, **\GTSi** This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$\boxed{A.\backslash\text{GTSincidence}/A.}$
 or $\boxed{A.\backslash\text{GTSinc}/A.}$
 or $\boxed{A.\backslash\text{GTSi}/A.}$
 will yield $\boxed{A.\downarrow A.}$.

$\boxed{v \backslash\text{GTSincidence}/ e}$
 or $\boxed{v \backslash\text{GTSinc}/ e}$
 or $\boxed{v \backslash\text{GTSi}/ e}$
 will yield $\boxed{v \downarrow e}$.

$\boxed{\$v \backslash\text{GTSincidence}/ e\$}$
 or $\boxed{\$v \backslash\text{GTSinc}/ e\$}$
 or $\boxed{\$v \backslash\text{GTSi}/ e\$}$
 will yield $\boxed{v \downarrow e}$.

\GTSincidenceRelation, \GTSincRel, \GTSiR This macro will draw an incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSincidenceRelation}/e\$$

or $\$v\backslash\text{GTSincRel}/e\$$

or $\$v\backslash\text{GTSiR}/e\$$

will yield $v \downarrow e$.

\GTSnegatedIncidence, \GTSnegInc, \GTSnI This macro will draw a negated incidence symbol.

$A.\backslash\text{GTSnegatedIncidence}/A.$

or $A.\backslash\text{GTSnegInc}/A.$

or $A.\backslash\text{GTSnI}/A.$

will yield $A.\not\downarrow A.$

$v \backslash\text{GTSnegatedIncidence}/ e$

or $v \backslash\text{GTSnegInc}/ e$

or $v \backslash\text{GTSnI}/ e$

will yield $v \not\downarrow e$.

$\$v \backslash\text{GTSnegatedIncidence}/ e\$$

or $\$v \backslash\text{GTSnegInc}/ e\$$

or $\$v \backslash\text{GTSnI}/ e\$$

will yield $v \not\downarrow e$.

\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR This macro will draw a negated incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSnegIncRel}/e\$$

or $\$v\backslash\text{GTSnIR}/e\$$

will yield $v \not\downarrow e$.

\GTSoutIncidence, \GTSoutInc, \GTSoI This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidence}/A.$

or $A.\backslash\text{GTSoutInc}/A.$

or $A.\backslash\text{GTSoI}/A.$

will yield $A.\uparrow A.$

$v \backslash\text{GTSoutIncidence}/ e$

or $v \backslash\text{GTSoutInc}/ e$

or $v \backslash\text{GTSoI}/ e$

will yield $v \uparrow e$.

$\$v \backslash\text{GTSoutIncidence}/ e\$$

or $\$v \backslash\text{GTSoutInc}/ e\$$

or $\$v \backslash\text{GTSoI}/ e\$$

will yield $v \uparrow e$.

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSoIR This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceRelation}/e\$$
or $\$v\backslash\text{GTSoutIncRel}/e\$$
or $\$v\backslash\text{GTSOI}/e\$$
will yield $v \dashv e$.

$\backslash\text{GTSnegatedOutIncidence}$, $\backslash\text{GTSnegOutInc}$, $\backslash\text{GTSnOI}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedOutIncidence}/A.$
or $A.\backslash\text{GTSnegOutInc}/A.$
or $A.\backslash\text{GTSnOI}/A.$
will yield $A.\nmid A.$.

$v \backslash\text{GTSnegatedOutIncidence}/ e$
or $v \backslash\text{GTSnegOutInc}/ e$
or $v \backslash\text{GTSnOI}/ e$
will yield $v \nmid e$.

$\$v \backslash\text{GTSnegatedOutIncidence}/ e\$$
or $\$v \backslash\text{GTSnegOutInc}/ e\$$
or $\$v \backslash\text{GTSnOI}/ e\$$
will yield $v \nmid e$.

$\backslash\text{GTSnegatedOutIncidenceRelation}$, $\backslash\text{GTSnegOutIncRel}$, $\backslash\text{GTSnOIR}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedOutIncidenceRelation}/e\$$
or $\$v\backslash\text{GTSnegOutIncRel}/e\$$
or $\$v\backslash\text{GTSnOIR}/e\$$
will yield $v \nmid e$.

$\backslash\text{GTSinIncidence}$, $\backslash\text{GTSinInc}$, $\backslash\text{GTSiI}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidence}/A.$
or $A.\backslash\text{GTSinInc}/A.$
or $A.\backslash\text{GTSiI}/A.$
will yield $A.\vdash A.$.

$v \backslash\text{GTSinIncidence}/ e$
or $v \backslash\text{GTSinInc}/ e$
or $v \backslash\text{GTSiI}/ e$
will yield $v \vdash e$.

$\$v \backslash\text{GTSinIncidence}/ e\$$
or $\$v \backslash\text{GTSinInc}/ e\$$
or $\$v \backslash\text{GTSiI}/ e\$$
will yield $v \vdash e$.

$\backslash\text{GTSinIncidenceRelation}$, $\backslash\text{GTSinIncRel}$, $\backslash\text{GTSiIR}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSinIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSinIncRel}/e\$$

or $\$v\backslash\text{GTSiIR}/e\$$

will yield $v \not\hookrightarrow e$.

$\backslash\text{GTSnegatedInIncidence}$, $\backslash\text{GTSnegInInc}$, $\backslash\text{GTSnII}$ This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedInIncidence}/A.$

or $A.\backslash\text{GTSnegInInc}/A.$

or $A.\backslash\text{GTSnII}/A.$

will yield $A.\not\hookrightarrow A.$

$v \backslash\text{GTSnegatedInIncidence}/ e$

or $v \backslash\text{GTSnegInInc}/ e$

or $v \backslash\text{GTSnII}/ e$

will yield $v \not\hookrightarrow e$.

$\$v \backslash\text{GTSnegatedInIncidence}/ e\$$

or $\$v \backslash\text{GTSnegInInc}/ e\$$

or $\$v \backslash\text{GTSnII}/ e\$$

will yield $v \not\hookrightarrow e$.

$\backslash\text{GTSnegatedInIncidenceRelation}$, $\backslash\text{GTSnegInIncRel}$, $\backslash\text{GTSnIIR}$ This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSnegInIncRel}/e\$$

or $\$v\backslash\text{GTSnIIR}/e\$$

will yield $v \not\hookrightarrow e$.

2.2.4 Monochrome adjacency macros

$\backslash\text{GTSadjacencyMonochrome}$, $\backslash\text{GTSadjMono}$, $\backslash\text{GTSadjMon}$, $\backslash\text{GTSaM}$ This macro will draw a monochrome adjacency symbol.

$A.\backslash\text{GTSadjacencyMonochrome}/A.$

or $A.\backslash\text{GTSadjMono}/A.$

or $A.\backslash\text{GTSadjMon}/A.$

or $A.\backslash\text{GTSaM}/A.$

will yield $A.\circ A.$

$u \backslash\text{GTSadjacencyMonochrome}/ v$

or $u \backslash\text{GTSadjMono}/ v$

or $u \backslash\text{GTSadjMon}/ v$

or $u \backslash\text{GTSaM}/ v$

will yield $u \circ v$.

$\$u \backslash\text{GTSadjacencyMonochrome}/ v\$$

or $\$u \backslash\text{GTSadjMono}/ v\$$

or $\$u \backslash\text{GTSadjMon}/ v\$$

or $\$u \backslash \text{GTSaM}/ v\$$
 will yield $u \circ v$.

$\backslash \text{GTSadjacencyMonochromeRelation}$, $\backslash \text{GTSadjMonoRel}$, $\backslash \text{GTSadjMonRel}$, $\backslash \text{GTSaMR}$
 This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

$\$u \backslash \text{GTSadjacencyMonochromeRelation}/ v\$$
 or $\$u \backslash \text{GTSadjMonoRel}/ v\$$
 or $\$u \backslash \text{GTSadjMonRel}/ v\$$
 or $\$u \backslash \text{GTSaMR}/ v\$$
 will yield $u \circ v$.

$\backslash \text{GTSunadjacencyMonochrome}$, $\backslash \text{GTSunadjMono}$, $\backslash \text{GTSunadjMon}$, $\backslash \text{GTSuM}$ This macro
 will draw a monochrome unadjacency symbol.

$A. \backslash \text{GTSunadjacencyMonochrome}/ A.$
 or $A. \backslash \text{GTSunadjMono}/ A.$
 or $A. \backslash \text{GTSunadjMon}/ A.$
 or $A. \backslash \text{GTSuM}/ A.$
 will yield $A. \circ A.$
 $u \backslash \text{GTSunadjacencyMonochrome}/ v$
 or $u \backslash \text{GTSunadjMono}/ v$
 or $u \backslash \text{GTSunadjMon}/ v$
 or $u \backslash \text{GTSuM}/ v$
 will yield $u \circ v$.
 $\$u \backslash \text{GTSunadjacencyMonochrome}/ v\$$
 or $\$u \backslash \text{GTSunadjMono}/ v\$$
 or $\$u \backslash \text{GTSunadjMon}/ v\$$
 or $\$u \backslash \text{GTSuM}/ v\$$
 will yield $u \circ v$.

$\backslash \text{GTSunadjacencyMonochromeRelation}$, $\backslash \text{GTSunadjMonoRel}$, $\backslash \text{GTSunadjMonRel}$, $\backslash \text{GTSuMR}$ This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

$\$u \backslash \text{GTSunadjacencyMonochromeRelation}/ v\$$
 or $\$u \backslash \text{GTSunadjMonoRel}/ v\$$
 or $\$u \backslash \text{GTSunadjMonRel}/ v\$$
 or $\$u \backslash \text{GTSuMR}/ v\$$
 will yield $u \circ v$.

$\backslash \text{GTSdirectedAdjacencyUpMonochrome}$, $\backslash \text{GTSdirAdjUpMono}$, $\backslash \text{GTSdirAdjUpMon}$, $\backslash \text{GTSdAUM}$ This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A. \backslash \text{GTSdirectedAdjacencyUpMonochrome}/ A.$
 or $A. \backslash \text{GTSdirAdjUpMono}/ A.$
 or $A. \backslash \text{GTSdirAdjUpMon}/ A.$
 or $A. \backslash \text{GTSdAUM}/ A.$
 will yield $A. \circ A.$.

`u \GTSdirectedAdjacencyUpMonochrome/ v`

or `u \GTSdirAdjUpMono/ v`

or `u \GTSdirAdjUpMon/ v`

or `u \GTSdAUM/ v`

will yield $u \overset{\circ}{\circ} v$.

`$u \GTSdirectedAdjacencyUpMonochrome/ v$`

or `$u \GTSdirAdjUpMono/ v$`

or `$u \GTSdirAdjUpMon/ v$`

or `$u \GTSdAUM/ v$`

will yield $u \overset{\circ}{\circ} v$.

`\GTSdirectedAdjacencyUpMonochromeRelation`, `\GTSdirAdjUpMonoRel`, `\GTSdirAdjUpMonRel`, `\GTSdAUMR` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

`$u\GTSdirectedAdjacencyUpMonochromeRelation/v$`

or `$u\GTSdirAdjUpMonoRel/v$`

or `$u\GTSdirAdjUpMonRel/v$`

or `$u\GTSdAUMR/v$`

will yield $u \overset{\circ}{\circ} v$.

`\GTSnegatedDirectedAdjacencyUpMonochrome`, `\GTSnegDirAdjUpMono`, `\GTSnegDirAdjUpMon`, `\GTSnDAUM` This macro will draw a monochrome negated directed adjacency upward symbol.

`A.\GTSnegatedDirectedAdjacencyUpMonochrome/A.`

or `A.\GTSnegDirAdjUpMono/A.`

or `A.\GTSnegDirAdjUpMon/A.`

or `A.\GTSnDAUM/A.`

will yield $A \overset{\circ}{\circ} A$.

`u \GTSnegatedDirectedAdjacencyUpMonochrome/ v`

or `u \GTSnegDirAdjUpMono/ v`

or `u \GTSnegDirAdjUpMon/ v`

or `u \GTSnDAUM/ v`

will yield $u \overset{\circ}{\circ} v$.

`$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v$`

or `$u \GTSnegDirAdjUpMono/ v$`

or `$u \GTSnegDirAdjUpMon/ v$`

or `$u \GTSnDAUM/ v$`

will yield $u \overset{\circ}{\circ} v$.

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation`, `\GTSnegDirAdjUpMonoRel`, `\GTSnegDirAdjUpMonRel`, `\GTSnDAUMR` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

`$u\GTSnegatedDirectedAdjacencyUpMonochromeRelation/v$`

or `$u\GTSnegDirAdjUpMonoRel/v$`

or `$u\GTSnegDirAdjUpMonRel/v$`

or $\$u\backslash\text{GTSnDAUMR}/v\$$
 will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSdirectedAdjacencyDownMonochrome}$, $\backslash\text{GTSdirAdjDownMono}$, $\backslash\text{GTSdirAdjDownMon}$, $\backslash\text{GTSdADM}$ This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$\text{A}.\backslash\text{GTSdirectedAdjacencyDownMonochrome}/\text{A}.$
 or $\text{A}.\backslash\text{GTSdirAdjDownMono}/\text{A}.$
 or $\text{A}.\backslash\text{GTSdirAdjDownMon}/\text{A}.$
 or $\text{A}.\backslash\text{GTSdADM}/\text{A}.$
 will yield $\text{A}.\overset{\circ}{\circ}\text{A}.$

$u \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v$
 or $u \backslash\text{GTSdirAdjDownMono}/ v$
 or $u \backslash\text{GTSdirAdjDownMon}/ v$
 or $u \backslash\text{GTSdADM}/ v$
 will yield $u \overset{\circ}{\circ} v$.

$\$u \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v\$$
 or $\$u \backslash\text{GTSdirAdjDownMono}/ v\$$
 or $\$u \backslash\text{GTSdirAdjDownMon}/ v\$$
 or $\$u \backslash\text{GTSdADM}/ v\$$
 will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}$, $\backslash\text{GTSdirAdjDownMonoRel}$, $\backslash\text{GTSdirAdjDownMonRel}$, $\backslash\text{GTSdADMR}$ This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}/v\$$
 or $\$u\backslash\text{GTSdirAdjDownMonoRel}/v\$$
 or $\$u\backslash\text{GTSdirAdjDownMonRel}/v\$$
 or $\$u\backslash\text{GTSdADMR}/v\$$
 will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}$, $\backslash\text{GTSnegDirAdjDownMono}$, $\backslash\text{GTSnegDirAdjDownMon}$, $\backslash\text{GTSnDADM}$ This macro will draw a monochrome negated directed adjacency downward symbol.

$\text{A}.\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/\text{A}.$
 or $\text{A}.\backslash\text{GTSnegDirAdjDownMono}/\text{A}.$
 or $\text{A}.\backslash\text{GTSnegDirAdjDownMon}/\text{A}.$
 or $\text{A}.\backslash\text{GTSnDADM}/\text{A}.$
 will yield $\text{A}.\overset{\circ}{\circ}\text{A}.$

$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v$
 or $u \backslash\text{GTSnegDirAdjDownMono}/ v$
 or $u \backslash\text{GTSnegDirAdjDownMon}/ v$
 or $u \backslash\text{GTSnDADM}/ v$
 will yield $u \overset{\circ}{\circ} v$.

`\GTSnegatedDirectedAdjacencyDownMonochrome/ v$`

or `\GTSnegDirAdjDownMono/ v$`

or `\GTSnegDirAdjDownMon/ v$`

or `\GTSnDADM/ v$`

will yield $u \overset{g}{\downarrow} v$.

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

`$u\GTSnegatedDirectedAdjacencyDownMonochromeRelation/v$`

or `$u\GTSnegDirAdjDownMonoRel/v$`

or `$u\GTSnegDirAdjDownMonRel/v$`

or `$u\GTSnDADMR/v$`

will yield $u \overset{g}{\downarrow} v$.

2.2.5 Monochrome incidence macros

`\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`A.\GTSincidenceMonochrome/A.`

or `A.\GTSincMono/A.`

or `A.\GTSincMon/A.`

or `A.\GTSiM/A.`

will yield $A \downarrow A$.

`v \GTSincidenceMonochrome/ e`

or `v \GTSincMono/ e`

or `v \GTSincMon/ e`

or `v \GTSiM/ e`

will yield $v \downarrow e$.

`$v \GTSincidenceMonochrome/ e$`

or `$v \GTSincMono/ e$`

or `$v \GTSincMon/ e$`

or `$v \GTSiM/ e$`

will yield $v \downarrow e$.

`\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

`$v\GTSincidenceMonochromeRelation/e$`

or `$v\GTSincMonoRel/e$`

or `$v\GTSincMonRel/e$`

or `$v\GTSiMR/e$`

will yield $v \downarrow e$.

`\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM` This macro will draw a monochrome negated incidence symbol.

`A.\GTSnegatedIncidenceMonochrome/A.`

or `A.\GTSnegIncMono/A.`

or $A.\backslash\text{GTSnegIncMon}/A.$

or $A.\backslash\text{GTSnIM}/A.$

will yield $A.\not\hookrightarrow A.$

$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e$

or $v \backslash\text{GTSnegIncMono}/ e$

or $v \backslash\text{GTSnegIncMon}/ e$

or $v \backslash\text{GTSnIM}/ e$

will yield $v \not\hookrightarrow e.$

$\$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSnegIncMono}/ e\$$

or $\$v \backslash\text{GTSnegIncMon}/ e\$$

or $\$v \backslash\text{GTSnIM}/ e\$$

will yield $v \not\hookrightarrow e.$

$\backslash\text{GTSnegatedIncidenceMonochromeRelation}$, $\backslash\text{GTSnegIncMonoRel}$, $\backslash\text{GTSnegIncMonRel}$, $\backslash\text{GTSnIMR}$ This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedIncidenceMonochromeRelation}/e\$$

or $\$v\backslash\text{GTSnegIncMonoRel}/e\$$

or $\$v\backslash\text{GTSnegIncMonRel}/e\$$

or $\$v\backslash\text{GTSnIMR}/e\$$

will yield $v \not\hookrightarrow e.$

$\backslash\text{GTSoutIncidenceMonochrome}$, $\backslash\text{GTSoutIncMono}$, $\backslash\text{GTSoutIncMon}$, $\backslash\text{GTSoutIM}$ This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidenceMonochrome}/A.$

or $A.\backslash\text{GTSoutIncMono}/A.$

or $A.\backslash\text{GTSoutIncMon}/A.$

or $A.\backslash\text{GTSoutIM}/A.$

will yield $A.\not\uparrow A.$

$v \backslash\text{GTSoutIncidenceMonochrome}/ e$

or $v \backslash\text{GTSoutIncMono}/ e$

or $v \backslash\text{GTSoutIncMon}/ e$

or $v \backslash\text{GTSoutIM}/ e$

will yield $v \not\uparrow e.$

$\$v \backslash\text{GTSoutIncidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSoutIncMono}/ e\$$

or $\$v \backslash\text{GTSoutIncMon}/ e\$$

or $\$v \backslash\text{GTSoutIM}/ e\$$

will yield $v \not\uparrow e.$

$\backslash\text{GTSoutIncidenceMonochromeRelation}$, $\backslash\text{GTSoutIncMonoRel}$, $\backslash\text{GTSoutIncMonRel}$, $\backslash\text{GTSoutIMR}$ This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceMonochromeRelation}/e\$$
or $\$v\backslash\text{GTSoutIncMonoRel}/e\$$
or $\$v\backslash\text{GTSoutIncMonRel}/e\$$
or $\$v\backslash\text{GTSOIMR}/e\$$
will yield $v \nrightarrow e$.

$\backslash\text{GTSnegatedOutIncidenceMonochrome}$, $\backslash\text{GTSnegOutIncMono}$, $\backslash\text{GTSnegOutIncMon}$, $\backslash\text{GTSnOIM}$ This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedOutIncidenceMonochrome}/A.$
or $A.\backslash\text{GTSnegOutIncMono}/A.$
or $A.\backslash\text{GTSnegOutIncMon}/A.$
or $A.\backslash\text{GTSnOIM}/A.$
will yield $A.\nrightarrow A.$.

$v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e$
or $v \backslash\text{GTSnegOutIncMono}/ e$
or $v \backslash\text{GTSnegOutIncMon}/ e$
or $v \backslash\text{GTSnOIM}/ e$
will yield $v \nrightarrow e$.

$\$v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSnegOutIncMono}/ e\$$
or $\$v \backslash\text{GTSnegOutIncMon}/ e\$$
or $\$v \backslash\text{GTSnOIM}/ e\$$
will yield $v \nrightarrow e$.

$\backslash\text{GTSnegatedOutIncidenceMonochromeRelation}$, $\backslash\text{GTSnegOutIncMonoRel}$, $\backslash\text{GTSnegOutIncMonRel}$, $\backslash\text{GTSnOIMR}$ This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedOutIncidenceMonochromeRelation}/e\$$
or $\$v\backslash\text{GTSnegOutIncMonoRel}/e\$$
or $\$v\backslash\text{GTSnegOutIncMonRel}/e\$$
or $\$v\backslash\text{GTSnOIMR}/e\$$
will yield $v \nrightarrow e$.

$\backslash\text{GTSinIncidenceMonochrome}$, $\backslash\text{GTSinIncMono}$, $\backslash\text{GTSinIncMon}$, $\backslash\text{GTSiIM}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidenceMonochrome}/A.$
or $A.\backslash\text{GTSinIncMono}/A.$
or $A.\backslash\text{GTSinIncMon}/A.$
or $A.\backslash\text{GTSiIM}/A.$
will yield $A.\rightarrow A.$.

$v \backslash\text{GTSinIncidenceMonochrome}/ e$
or $v \backslash\text{GTSinIncMono}/ e$
or $v \backslash\text{GTSinIncMon}/ e$

or $\boxed{v \setminus \text{GTSiIM} / e}$

will yield $\boxed{v \nmid e}$.

$\boxed{\$v \setminus \text{GTSinIncidenceMonochrome} / e\$}$

or $\boxed{\$v \setminus \text{GTSinIncMono} / e\$}$

or $\boxed{\$v \setminus \text{GTSinIncMon} / e\$}$

or $\boxed{\$v \setminus \text{GTSiIM} / e\$}$

will yield $\boxed{v \nmid e}$.

$\setminus \text{GTSinIncidenceMonochromeRelation}$, $\setminus \text{GTSinIncMonoRel}$, $\setminus \text{GTSinIncMonRel}$, $\setminus \text{GTSiIMR}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\boxed{\$v \setminus \text{GTSinIncidenceMonochromeRelation} / e\$}$

or $\boxed{\$v \setminus \text{GTSinIncMonoRel} / e\$}$

or $\boxed{\$v \setminus \text{GTSinIncMonRel} / e\$}$

or $\boxed{\$v \setminus \text{GTSiIMR} / e\$}$

will yield $\boxed{v \nmid e}$.

$\setminus \text{GTSnegatedInIncidenceMonochrome}$, $\setminus \text{GTSnegInIncMono}$, $\setminus \text{GTSnegInIncMon}$, $\setminus \text{GTSniIM}$ This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

$\boxed{A. \setminus \text{GTSnegatedInIncidenceMonochrome} / A.}$

or $\boxed{A. \setminus \text{GTSnegInIncMono} / A.}$

or $\boxed{A. \setminus \text{GTSnegInIncMon} / A.}$

or $\boxed{A. \setminus \text{GTSniIM} / A.}$

will yield $\boxed{A. \nmid A.}$.

$\boxed{v \setminus \text{GTSnegatedInIncidenceMonochrome} / e}$

or $\boxed{v \setminus \text{GTSnegInIncMono} / e}$

or $\boxed{v \setminus \text{GTSnegInIncMon} / e}$

or $\boxed{v \setminus \text{GTSniIM} / e}$

will yield $\boxed{v \nmid e}$.

$\boxed{\$v \setminus \text{GTSnegatedInIncidenceMonochrome} / e\$}$

or $\boxed{\$v \setminus \text{GTSnegInIncMono} / e\$}$

or $\boxed{\$v \setminus \text{GTSnegInIncMon} / e\$}$

or $\boxed{\$v \setminus \text{GTSniIM} / e\$}$

will yield $\boxed{v \nmid e}$.

$\setminus \text{GTSnegatedInIncidenceMonochromeRelation}$, $\setminus \text{GTSnegInIncMonoRel}$, $\setminus \text{GTSnegInIncMonRel}$, $\setminus \text{GTSniIMR}$ This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\boxed{\$v \setminus \text{GTSnegatedInIncidenceMonochromeRelation} / e\$}$

or $\boxed{\$v \setminus \text{GTSnegInIncMonoRel} / e\$}$

or $\boxed{\$v \setminus \text{GTSnegInIncMonRel} / e\$}$

or $\boxed{\$v \setminus \text{GTSniIMR} / e\$}$

will yield $\boxed{v \nmid e}$.

2.2.6 Meta-logic macros

\GTSMetaNot, \GTSMN This macro will draw a meta-not or relation-not symbol.

A.\GTSMetaNot/A.

or **A.\GTSMN/A.**

will yield $A.\boxminus A.$

u \GTSMetaNot/\GTSDAU/ v

or **u \GTSMN/\GTSDAU/ v**

will yield $u \boxminus v.$

\$u \GTSMetaNot/ \GTSDAU/ v\$

or **\$u \GTSMN/ \GTSDAU/ v\$**

will yield $u \boxminus v.$

\GTSMetaAnd, \GTSMa This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

A.\GTSMetaAnd/A.

or **A.\GTSMa/A.**

will yield $A.\boxplus A.$

u \GTSDAD/\GTSMetaAnd/\GTSDAU/ v

or **u \GTSDAD/\GTSMa/\GTSDAU/ v**

will yield $u \boxplus v.$

\$u \GTSDAD/ \GTSMetaAnd/ \GTSDAU/ v\$

or **\$u \GTSDAD/ \GTSMa/ \GTSDAU/ v\$**

will yield $u \boxplus v.$

\GTSMetaAndRelation, \GTSMetaAndRel, \GTSMAR This macro will draw a meta-and symbol with relation spacing in math mode.

\$u \GTSDAD/\GTSMetaAndRelation/\GTSDAU/ v\$

or **\$u \GTSDAD/\GTSMetaAndRel/\GTSDAU/ v\$**

or **\$u \GTSDAD/\GTSMAR/\GTSDAU/ v\$**

will yield $u \boxplus v.$

\GTSMetaOr, \GTSMo This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

A.\GTSMetaOr/A.

or **A.\GTSMo/A.**

will yield $A.\boxdot A.$

u \GTSDAD/\GTSMetaOr/\GTSDAU/ v

or **u \GTSDAD/\GTSMo/\GTSDAU/ v**

will yield $u \boxdot v.$

\$u \GTSDAD/ \GTSMetaOr/ \GTSDAU/ v\$

or **\$u \GTSDAD/ \GTSMo/ \GTSDAU/ v\$**

will yield $u \boxdot v.$

\GTSMetaOrRelation, \GTSMetaOrRel, \GTSMOR This macro will draw a meta-or symbol with relation spacing in math mode.

\$u \GTSDAD/\GTSMetaOrRelation/\GTSDAU/ v\$

or **\$u \GTSDAD/\GTSMetaOrRel/\GTSDAU/ v\$**

or **\$u \GTSDAD/\GTSMOR/\GTSDAU/ v\$**

will yield $u \boxdot v.$

23

$$\forall u, v, u \text{ } \begin{array}{c} \circ \\ \downarrow \\ \circ \end{array} \begin{array}{c} \circ \\ \downarrow \\ \circ \end{array} \overset{\text{m}}{\vee} \begin{array}{c} \circ \\ \downarrow \\ \circ \end{array} \begin{array}{c} \circ \\ \downarrow \\ \circ \end{array} v.$$
$$\forall u, v, u \left[\begin{array}{c} \textcircled{\text{m}} \textcircled{\text{m}} \textcircled{\text{m}} \textcircled{\text{m}} \\ \textcircled{\text{V}} \textcircled{\text{V}} \textcircled{\text{V}} \textcircled{\text{V}} \end{array} \right] v.$$
$$\forall u, v, u \left[\begin{array}{c} \textcircled{\text{v}} \textcircled{\text{u}} \textcircled{\text{v}} \\ \textcircled{\text{v}} \textcircled{\text{u}} \textcircled{\text{v}} \\ \textcircled{\text{v}} \textcircled{\text{u}} \textcircled{\text{v}} \end{array} \right] v.$$
$$\exists u, v, u \left[\overset{\text{m}}{\underset{\text{V}}{\text{V}}} \begin{array}{c} \text{red} \\ \text{green} \\ \text{blue} \end{array} \right] v.$$

```

1 \package)
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17

```

Arguments:

$\{\langle commandcode \rangle\}$ The code of the command to create.

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

`\{commandname\}` The name of the command to create.

```

21 \ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
22   \@ifdefinable{#1}{\protected\def#1/{#2}}
23 }}
```

Argument:

```

24 \newcommand{\GTSsetMainColor}[1]{%
25 \edef\GTS@mainColorTemp{#1}%
26 \edef\GTS@mainColorBackup{\GTS@mainColor}%
27 \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }

```

Argument:

24

```

29 \newcommand{\GTSsetSecondColor}[1]{%
30 \edef\GTS@secondColorTemp{#1}%
31 \edef\GTS@secondColorBackup{\GTS@secondColor}%
32 \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }

```

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and **\GTS@b**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

{<mainColor>} If non-empty, this color will be used as main color.
{<secondColor>} If non-empty, this color will be used as second color.
{<drawSlash>} Non-empty implies that a negation will be drawn.
{<drawUpArrow>} Non-empty implies that an up-arrow will be drawn.
{<drawDownArrow>} Non-empty implies that a down-arrow will be drawn.

```

34 \newcommand{\GTS@a}[5]{%
35 \if#1\empty\else%
36 \GTSsetMainColor{#1}%
37 \fi%
38 \if#2\empty\else%
39 \GTSsetSecondColor{#2}%
40 \fi%
41 \text{\ensuremath{%
42 \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43 %
44 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45 $\mathcolor{\GTS@secondColor}{\circ}$%
46 }}}}
47 \hspace{-0.39ex}%
48 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}}}
51 \hspace{-0.12ex}%
52 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54 }}}}
55 \hspace{-0.40ex}%
56 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57 $\mathcolor{\GTS@secondColor}{\circ}$%
58 }}}}
59 \hspace{-0.39ex}%
60 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61 $\mathcolor{\GTS@secondColor}{\bullet}$%
62 }}}}
63 \hspace{0.27ex}%

```

if **\GTS@drawUpArrow** empty else

```

64 \if#4\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67 \textcolor{\GTS@mainColor}{\textasciicircum}%
68 }}}}
69 \fi%

```

if **\GTS@drawDownArrow** empty else

```

70 \if#5\empty\else%
71 \hspace{-0.65ex}%
72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}}}
75 \fi%

```

if **\GTS@drawSlash** empty else

```

76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}}}

```

```

81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

{<mainColor>} If non-empty, this color will be used as main color.
{<secondColor>} If non-empty, this color will be used as second color.
{<drawSlash>} Non-empty implies that a negation will be drawn.
{<drawUpArrow>} Non-empty implies that an up-arrow will be drawn.
{<drawDownArrow>} Non-empty implies that a down-arrow will be drawn.
{<relation>} Non-empty implies that relation spacing will be added.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%
96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

\GTSxadjacency, **\GTSxadj**, **\GTSxa** Versatile macros for adjacency that can be given options.

Argument:

[<optionsList>] Sets the following options if not empty:

mainColor=color1 or **mainC=**color1 or **mColor=**color1 or **mC=**color1,
secondColor=color2 or **secondC=**color2 or **sColor=**color2 or **sC=**color2,
monochrome or **mono** or **mon** or **m** (equivalent to **mC=black**, **sC=black**),
negated or **not** or **n**,
up or **u**,
down or **d**,
relation or **rel** or **r**.

```

99 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{GTSxa}{monochrome}[1]%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{GTSxa}{mono}[1]%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{GTSxa}{mon}[1]%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{GTSxa}{m}[1]%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}

```

```

123 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1] [] {%
127 \def\GTS@@mainColor{}%
128 \def\GTS@@secondColor{}%
129 \def\GTS@@drawSlash{}%
130 \def\GTS@@drawUpArrow{}%
131 \def\GTS@@drawDownArrow{}%
132 \def\GTS@@relation{}%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
139
140 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

\GTSadjacency, \GTSadj, \GTSa This macro will draw an adjacency symbol.

```

141 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
142 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
143 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

\GTSadjacencyRelation, \GTSadjRel, \GTSaR This macro will draw an adjacency symbol with relation spacing in math mode.

```

144 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
145 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
146 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

\GTSunadjacency, \GTSunadj, \GTSu This macro will draw an unadjacency symbol.

```

147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}
148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}
149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}

```

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR This macro will draw an unadjacency symbol with relation spacing in math mode.

```

150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}{}}
154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}{}}
155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}{}}

```

\GTSdirectedAdjacencyUpRelation, \GTSdirAdjUpRel, \GTSdAUR This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

\GTSnegatedDirectedAdjacencyUp, \GTSnegDirAdjUp, \GTSnDAU This macro will draw a negated directed adjacency upward symbol.

```

159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}{}}
160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}{}}
161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}{}}

```

\GTSnegatedDirectedAdjacencyUpRelation, \GTSnegDirAdjUpRel, \GTSnDAUR This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
163 \mathrel{\GTSnDAU/}}%

```

```

164 }
165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

\GTSdirectedAdjacencyDown, \GTSdirAdjDown, \GTSdAD This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}}
168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}}
169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}}

```

\GTSdirectedAdjacencyDownRelation, \GTSdirAdjDownRel, \GTSdADR This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

\GTSnegatedDirectedAdjacencyDown, \GTSnegDirAdjDown, \GTSnDAD This macro will draw a negated directed adjacency downward symbol.

```

173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{1}}
174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{1}}
175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{1}}

```

\GTSnegatedDirectedAdjacencyDownRelation, \GTSnegDirAdjDownRel, \GTSnDADR This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
177 \mathrel{\GTSnDAD/}%
178 }
179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

\GTS@i This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

{<mainColor>} If non-empty, this color will be used as main color.
{<secondColor>} If non-empty, this color will be used as second color.
{<drawSlash>} Non-empty implies that a negation will be drawn.
{<drawUpArrow>} Non-empty implies that an up-arrow will be drawn.
{<drawDownArrow>} Non-empty implies that a down-arrow will be drawn.

```

181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}%
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}%
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }}%
202 \hspace{-0.18ex}%

```

```

203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }}%
206 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }}%
212 \fi%
if GTS@drawDownArrow empty else
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }}%
218 \fi%
if GTS@drawSlash empty else
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%
223 }}%
224 \fi%
225 %
226 }}%
227 }}%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and \GTS@i: \GTSxincidence, \GTSxinc, and \GTSxi.

Arguments:

- {*mainColor*}} If non-empty, this color will be used as main color.
- {*secondColor*}} If non-empty, this color will be used as second color.
- {*drawSlash*}} Non-empty implies that a negation will be drawn.
- {*drawUpArrow*}} Non-empty implies that an up-arrow will be drawn.
- {*drawDownArrow*}} Non-empty implies that a down-arrow will be drawn.
- {*relation*}} Non-empty implies that relation spacing will be added.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxincidence, \GTSxinc, \GTSxi Versatile macros for incidence that can be given options.

Argument:

[*optionsList*] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

```

242 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
244 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
245 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
246 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
248 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
249 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
250 \define@key{GTSxi}{monochrome}[1]%
251 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
252 \define@key{GTSxi}{mono}[1]%
253 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
254 \define@key{GTSxi}{mon}[1]%
255 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
256 \define@key{GTSxi}{m}[1]%
257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][]{%
270 \def\GTS@@mainColor{}%
271 \def\GTS@@secondColor{}%
272 \def\GTS@@drawSlash{}%
273 \def\GTS@@drawUpArrow{}%
274 \def\GTS@@drawDownArrow{}%
275 \def\GTS@@relation{}%
276 \setkeys{GTSxi}{#1}%
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][]{\GTSxi[#1]}
282
283 \newcommand{\GTSxincidence}[1][]{\GTSxi[#1]}

```

\GTSincidence, \GTSinc, \GTSi This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

284 \GTS@newcommand{\GTSincidence}{\GTSi{}{}{}{}}
285 \GTS@newcommand{\GTSinc}{\GTSi{}{}{}{}}
286 \GTS@newcommand{\GTSi}{\GTSi{}{}{}{}}

```

\GTSincidenceRelation, \GTSincRel, \GTSiR This macro will draw an incidence symbol with relation spacing in math mode.

```

287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

```

\GTSnegatedIncidence, \GTSnegInc, \GTSnI This macro will draw a negated incidence symbol.

```

290 \GTS@newcommand{\GTSnegatedIncidence}{\GTSi{}{}{1}{}}
291 \GTS@newcommand{\GTSnegInc}{\GTSi{}{}{1}{}}
292 \GTS@newcommand{\GTSnI}{\GTSi{}{}{1}{}}

```

\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR This macro will draw a negated incidence symbol with relation spacing in math mode.

```

293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}

```

294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}

295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

\GTSoutIncidence, \GTSoutInc, \GTSOI This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{}{1}{}}

297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{}{1}{}}

298 \GTS@newcommand{\GTSOI}{\GTS@i{}{}{}{1}{}}

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSOIR This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}

300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}

301 \GTS@newcommand{\GTSOIR}{\mathrel{\GTSOI/}}

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}}

303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}}

304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}}

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}

306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}

307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

\GTSinIncidence, \GTSinInc, \GTSiI This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}{}}

309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}{}}

310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}{}}

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}

312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}

313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{1}{}}

315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{1}{}}

316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{1}{}}

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}

318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}

319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM This macro will draw a monochrome adjacency symbol.

320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}}{}}

321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}}{}}

322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}}{}}

323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}}{}}

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}

325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}

326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}

327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM This macro will draw a monochrome unadjacency symbol.

```
328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}}
329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}}
330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}}
```

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```
332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}
```

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
337 \GTS@a{black}{black}{1}{}}
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{1}{}}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{1}{}}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{1}{}}
```

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```
342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
343 \mathrel{\GTSdAUM/}}
344 }
345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}
```

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

```
348 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
349 \GTS@a{black}{black}{1}{1}{}}
350 }
351 \GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
352 \GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
353 \GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}
```

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```
354 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
355 \mathrel{\GTSnDAUM/}}
356 }
357 \GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
358 \GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
359 \GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}
```

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
360 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
361 \GTS@a{black}{black}{1}{}}
362 }
363 \GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{1}{}}
364 \GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{1}{}}
365 \GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{1}{}}
```

\GTSdirectedAdjacencyDownMonochromeRelation, \GTSdirAdjDownMonoRel, \GTSdirAdjDownMonRel, \GTSdADMR This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```
366 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
367 \mathrel{\GTSdADM/}%
368 }
369 \GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
370 \GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
371 \GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}
```

\GTSnegatedDirectedAdjacencyDownMonochrome, \GTSnegDirAdjDownMono, \GTSnegDirAdjDownMon, \GTSnDADM This macro will draw a monochrome negated directed adjacency downward symbol.

```
372 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
373 \GTS@a{black}{black}{1}{1}%
374 }
375 \GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{1}}
376 \GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{1}}
377 \GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{1}}
```

\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```
378 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
379 \mathrel{\GTSnDADM/}%
380 }
381 \GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
382 \GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
383 \GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}
```

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
384 \GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{}}
385 \GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{}}
386 \GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{}}
387 \GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{}}
```

\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```
388 \GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}
389 \GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}
390 \GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}
391 \GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}
```

\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM This macro will draw a monochrome negated incidence symbol.

```
392 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
393 \GTS@i{black}{black}{1}}
394 }
395 \GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}}
396 \GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}}
397 \GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}}
```

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```
398 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
399 \GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
400 \GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
401 \GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}
```

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSOIM This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
402 \GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{}}{1}{1}}
403 \GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{}}{1}{1}}
404 \GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{}}{1}{1}}
405 \GTS@newcommand{\GTSOIM}{\GTS@i{black}{black}{}}{1}{1}}
```

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSOIMR This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
406 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSOIM/}}
407 \GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSOIM/}}
408 \GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSOIM/}}
409 \GTS@newcommand{\GTSOIMR}{\mathrel{\GTSOIM/}}
```

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GTSnOIM This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```
410 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
411 \GTS@i{black}{black}{1}{1}}{1}{1}}
412 }
413 \GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}}{1}{1}}
414 \GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}}{1}{1}}
415 \GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}}{1}{1}}
```

\GTSnegatedOutIncidenceMonochromeRelation, \GTSnegOutIncMonoRel, \GTSnegOutIncMonRel, \GTSnOIMR This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
416 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
417 \mathrel{\GTSnOIM/}}{1}{1}}
418 }
419 \GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
420 \GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
421 \GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}
```

\GTSinIncidenceMonochrome, \GTSinIncMono, \GTSinIncMon, \GTSiIM This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
422 \GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{}}{1}{1}}
423 \GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{}}{1}{1}}
424 \GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{}}{1}{1}}
425 \GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{}}{1}{1}}
```

\GTSinIncidenceMonochromeRelation, \GTSinIncMonoRel, \GTSinIncMonRel, \GTSiIMR This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
426 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
427 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
428 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
429 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}
```

\GTSnegatedInIncidenceMonochrome, \GTSnegInIncMono, \GTSnegInIncMon, \GTSnIIM This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```
430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
431 \GTS@i{black}{black}{1}{1}}{1}{1}}
432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}{1}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}{1}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}{1}{1}}
```

\GTSnegatedInIncidenceMonochromeRelation, \GTSnegInIncMonoRel, \GTSnegInIncMonRel, \GTSnIIMR This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
437 \mathrel{\GTSnIIM/}}%
438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}
```

\GTSmetaNot, \GTSmN This macro will draw a meta-not or relation-not symbol.

```
442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}[1]{%
445 \raisebox{0ex}{\scalebox{1}[1]{\not$}}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}}%
448 \hspace{0.2ex}%
449 }}}%
450 }}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}
```

\GTSmetaAnd, \GTSmA This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

```
453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
456 \raisebox{0ex}{\scalebox{1}[1]{\wedge$}}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
459 }}}%
460 }}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}
```

\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR This macro will draw a meta-and symbol with relation spacing in math mode.

```
463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}
```

\GTSmetaOr, \GTSmO This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol.

```
466 \GTS@newprotectedcommand{\GTSmO}{%
467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\vee$}}}%
470 \hspace{-1.46ex}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}}%
473 }}%
474 }
475 \GTS@newcommand{\GTSmetaOr}{\GTSmO/}
```

\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR This macro will draw a meta-or symbol with relation spacing in math mode.

```
476 \GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
```

```

477 \GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}

```

\GTSxmetallogic, \GTSxml Versatile macros for meta-logic symbols that can be given options.

Argument:

[*\langle optionsList \rangle*] Sets the following options if not empty:

not or n,
and or a,
or or o,
relation or rel or r.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

```

479 \define@key{GTSxml}{not}[1]{\def\GTS@@drawMetaNot{1}}
480 \define@key{GTSxml}{n}[1]{\def\GTS@@drawMetaNot{1}}
481 \define@key{GTSxml}{and}[1]{\def\GTS@@drawMetaAnd{1}}
482 \define@key{GTSxml}{a}[1]{\def\GTS@@drawMetaAnd{1}}
483 \define@key{GTSxml}{or}[1]{\def\GTS@@drawMetaOr{1}}
484 \define@key{GTSxml}{o}[1]{\def\GTS@@drawMetaOr{1}}
485 \define@key{GTSxml}{relation}[1]{\def\GTS@@relation{1}}
486 \define@key{GTSxml}{rel}[1]{\def\GTS@@relation{1}}
487 \define@key{GTSxml}{r}[1]{\def\GTS@@relation{1}}
488
489 \newcommand{\GTSxml}[1][{}]{%
490 \def\GTS@@drawMetaNot{}%
491 \def\GTS@@drawMetaAnd{}%
492 \def\GTS@@drawMetaOr{}%
493 \def\GTS@@relation{}%
494 \setkeys{GTSxml}{#1}%
495 \if\GTS@@drawMetaNot1%
496 \GTSmN/%
497 \else\if\GTS@@drawMetaAnd1%
498 \if\GTS@@relation{1}%
499 \GTSmAR/%
500 \else%
501 \GTSmA/%
502 \fi%
503 \else\if\GTS@@drawMetaOr1%
504 \if\GTS@@relation{1}%
505 \GTSmOR/%
506 \else%
507 \GTSmO/%
508 \fi%
509 \else%
510 Wrong call to \textbackslash{GTSxml}%
511 \fi%
512 \fi%
513 \fi%
514 }
515
516 \newcommand{\GTSxmetallogic}[1][{}]{\GTSxml[#1]}

```

\GTSx One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

[*\langle optionsList \rangle*] Sets the following options if not empty:

- If *\langle symbolType \rangle* below is a, see \GTSxa:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

- If $\{\langle symbolType \rangle\}$ below is *i*, see `\GTSxi`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`in` or `i`,
`out` or `o`,
`relation` or `rel` or `r`.
- If $\{\langle symbolType \rangle\}$ below is *m*, see `\GTSxml`:
`not` or `n`,
`and` or `a`,
`or` or `o`,
`relation` or `rel` or `r`.

$\{\langle symbolType \rangle\}$ a mandatory switch; it can take value *a* for adjacency, *i* for incidence, *m* for meta-logic.

```

517 \newcommand{\GTSx}[2][]{%
518 \if#2a%
519 \GTSxa[#1]%
520 \else\if#2i%
521 \GTSxi[#1]%
522 \else\if#2m%
523 \GTSxml[#1]%
524 \else%
525 Wrong call to \textbackslash{}GTSx%
526 \fi%
527 \fi%
528 \fi%
529 }

530 \endinput
531 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 FAQ

Question: Why `GraphTheorySymbol` instead of `GraphTheorySymbols`? `GraphTheorySymbol` does contain many symbols.

Answer: Because I followed one common usage on CTAN.

Here is the list of all current packages with the word "symbol" in their name:

`actuarialsymbol`, `adfsymbols`, `compact-symbols`, `cookingsymbols`, `FdSymbol`, `jpnedumathsymbols`, `KKsymbols`, `listofsymbols`, `losymbol`, `maths-symbols`, `mdsymbol`, `MnSymbol`, `open-everyday-symbols`, `qsymbols`, `SVRsymbols`, `symbolindex`, `The Comprehensive LaTeX Symbol List`, `tikz-cookingsymbols`, `tikzsymbols`.

7 "symbol" plus `GraphTheorySymbol` 8, 12 "symbols".

"symb" abbreviation, "sym", "sy" and "s" are also quite common;

`oPlotSymbl` did it its own way ;) XD.

Because of the goal of `GraphTheorySymbol`, I was mainly looking at packages like `FdSymbol`, `mdsymbol`, `MnSymbol`, and `The Comprehensive LaTeX Symbol List` before creating my package; hence I followed this apparent majority, instead of listening to my thought that `Symbols` would be better.

After the package was uploaded to CTAN for a version update, this question came again, but CTAN admins were reluctant to change the package name afterward.

Otherwise, it is a matter of hours to modify and upload `GraphTheorySymbols` with version 2.0.0 to CTAN.

6 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

7 Change History

v0.1.0 – 2026-08-01	operations on relations	1
General: Initial version		1
v1.0.0 – 2026-08-26	General: One macro to rule them all	1
General: First public release		1
v1.1.0 – 2026-08-29	General: Use noprint option of doc package to	
General: Added a memo	avoid too big left margins, and use	
v1.1.1 – 2026-08-30	paragraph command instead	1
General: Small corrections and improvements		1
v1.2.0 – 2026-09-06	v1.5.2 – 2026-09-14	
General: CTAN URL, new macros for same	General: Add A4, USLetter, and Book	
set of symbols	variants of PDFs.	1
v1.2.1 – 2026-09-08	v1.6.0 – 2026-09-18	
General: Smaller tests output	General: Add verbatim code for compound	
v1.3.0 – 2026-09-08	adjacency in memo.	1
General: New macros for monochrome	v1.6.1 – 2026-09-19	
symbols	General: Rename test(.tex, etc.) to	
v1.4.0 – 2026-09-12	graph-theory-symbol-test(.tex, etc.).	1
General: New macros for symbols for logical	v1.7.0 – 2026-09-21	
	General: Add FAQ section.	1

Index

G

<code>\GTS@@drawDownArrow</code>	120, 121, 131, 135, 263, 264, 274, 278
<code>\GTS@@drawMetaAnd</code>	481, 482, 491, 497
<code>\GTS@@drawMetaNot</code>	479, 480, 490, 495
<code>\GTS@@drawMetaOr</code>	483, 484, 492, 503
<code>\GTS@@drawSlash</code>	115, 116, 117, 129, 134, 258, 259, 260, 272, 277
<code>\GTS@@drawUpArrow</code>	118, 119, 130, 135, 261, 262, 273, 278
<code>\GTS@mainColor</code> 99, 100, 101, 102, 108, 110, 112, 114, 127, 134, 242, 243, 244, 245, 251, 253, 255, 257, 270, 277	
<code>\GTS@@relation</code>	122, 123, 124, 132, 135, 265, 266, 267, 275, 278, 485, 486, 487, 493, 498, 504
<code>\GTS@secondColor</code> . 103, 104, 105, 106, 108, 110, 112, 114, 128, 134, 246, 247, 248, 249, 251, 253, 255, 257, 271, 277	
<code>\GTS@a</code> . 7, 34, 94, 96, 141, 142, 143, 147, 148, 149, 153, 154, 155, 159, 160, 161, 167, 168, 169, 173, 174, 175, 320, 321, 322, 323, 328, 329, 330, 331, 337, 339, 340, 341, 349, 351, 352, 353, 361, 363, 364, 365, 373, 375, 376, 377	
<code>\GTS@b</code>	7, 92, 134
<code>\GTS@drawDownArrow</code>	16
<code>\GTS@drawSlash</code>	14
<code>\GTS@drawUpArrow</code>	15
<code>\GTS@i</code> . 11, 181, 237, 239, 284, 285, 286, 290, 291, 292, 296, 297, 298, 302, 303, 304, 308, 309, 310, 314, 315, 316, 384, 385, 386, 387, 393, 395, 396, 397, 402, 403, 404, 405, 411, 413, 414, 415, 422, 423, 424, 425, 431, 433, 434, 435	
<code>\GTS@j</code>	11, 235, 277
<code>\GTS@mainColor</code>	10, 26, 27, 53, 67, 73, 204
<code>\GTS@mainColorBackup</code>	12, 26, 86, 229
<code>\GTS@mainColorTemp</code>	25, 27
<code>\GTS@newcommand</code> . . 7, 18, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 179, 180, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 339, 340, 341, 342, 345, 346, 347, 348, 351, 352, 353, 354, 357, 358, 359, 360, 363, 364, 365, 366, 369, 370, 371, 372, 375, 376, 377, 378, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 413, 414, 415, 416, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 433, 434, 435, 436, 439, 440, 441, 452, 462, 463, 464, 465, 475, 476, 477, 478	
<code>\GTS@newprotectedcommand</code>	7, 21, 442, 453, 466
<code>\GTS@secondColor</code>	11, 31, 32, 45, 49, 57, 61, 192, 196, 200, 210, 216

\GTS@secondColorBackup	13, 31, 89, 232
\GTS@secondColorTemp	30, 32
\GTSa	8, <u>141</u> , 144, 145, 146
\GTSadj	8, <u>141</u>
\GTSadjacency	8, <u>141</u>
\GTSadjacencyMonochrome	14, <u>320</u>
\GTSadjacencyMonochromeRelation	15, <u>324</u>
\GTSadjacencyRelation	8, <u>144</u>
\GTSadjMon	14, <u>320</u>
\GTSadjMono	14, <u>320</u>
\GTSadjMonoRel	15, <u>324</u>
\GTSadjMonRel	15, <u>324</u>
\GTSadjRel	8, <u>144</u>
\GTSaM	14, <u>320</u> , 324, 325, 326, 327
\GTSaMR	15, <u>324</u>
\GTSaR	8, <u>144</u>
\GTSdAD	10, <u>167</u> , 170, 171, 172
\GTSdADM	17, <u>360</u> , 367, 369, 370, 371
\GTSdADMR	17, <u>366</u>
\GTSdADR	10, <u>170</u>
\GTSdAU	9, <u>153</u> , 156, 157, 158
\GTSdAUM	15, <u>336</u> , 343, 345, 346, 347
\GTSdAUMR	16, <u>342</u>
\GTSdAUR	9, <u>156</u>
\GTSdirAdjDown	10, <u>167</u>
\GTSdirAdjDownMon	17, <u>360</u>
\GTSdirAdjDownMono	17, <u>360</u>
\GTSdirAdjDownMonoRel	17, <u>366</u>
\GTSdirAdjDownMonRel	17, <u>366</u>
\GTSdirAdjDownRel	10, <u>170</u>
\GTSdirAdjUp	9, <u>153</u>
\GTSdirAdjUpMon	15, <u>336</u>
\GTSdirAdjUpMono	15, <u>336</u>
\GTSdirAdjUpMonoRel	16, <u>342</u>
\GTSdirAdjUpMonRel	16, <u>342</u>
\GTSdirAdjUpRel	9, <u>156</u>
\GTSdirectedAdjacencyDown	10, <u>167</u>
\GTSdirectedAdjacencyDownMonochrome	17, <u>360</u>
\GTSdirectedAdjacencyDownMonochromeRelation	17, <u>366</u>
\GTSdirectedAdjacencyDownRelation	10, <u>170</u>
\GTSdirectedAdjacencyUp	9, <u>153</u>
\GTSdirectedAdjacencyUpMonochrome	15, <u>336</u>
\GTSdirectedAdjacencyUpMonochromeRelation	16, <u>342</u>
\GTSdirectedAdjacencyUpRelation	9, <u>156</u>
\GTSi	11, <u>284</u> , 287, 288, 289
\GTSiI	13, <u>308</u> , 311, 312, 313
\GTSiIM	20, <u>422</u> , 426, 427, 428, 429
\GTSiIMR	21, <u>426</u>
\GTSiIR	13, <u>311</u>
\GTSiM	18, <u>384</u> , 388, 389, 390, 391
\GTSiMR	18, <u>388</u>
\GTSinc	11, <u>284</u>
\GTSincidence	11, <u>284</u>
\GTSincidenceMonochrome	18, <u>384</u>
\GTSincidenceMonochromeRelation	18, <u>388</u>
\GTSincidenceRelation	12, <u>287</u>
\GTSincMon	18, <u>384</u>
\GTSincMono	18, <u>384</u>
\GTSincMonoRel	18, <u>388</u>
\GTSincMonRel	18, <u>388</u>
\GTSincRel	12, <u>287</u>

\GTSinInc	13, 308
\GTSinIncidence	13, 308
\GTSinIncidenceMonochrome	20, 422
\GTSinIncidenceMonochromeRelation	21, 426
\GTSinIncidenceRelation	13, 311
\GTSinIncMon	20, 422
\GTSinIncMono	20, 422
\GTSinIncMonoRel	21, 426
\GTSinIncMonRel	21, 426
\GTSinIncRel	13, 311
\GTSiR	12, 287
\GTSmA	22, 453, 463, 464, 465, 501
\GTSmAR	22, 463, 499
\GTSmetaAnd	22, 453
\GTSmetaAndRel	22, 463
\GTSmetaAndRelation	22, 463
\GTSmetaNot	22, 442
\GTSmetaOr	22, 466
\GTSmetaOrRel	22, 476
\GTSmetaOrRelation	22, 476
\GTSmN	22, 442, 496
\GTSmO	22, 466, 476, 477, 478, 507
\GTSmOR	22, 476, 505
\GTSnDAD	10, 173, 177, 179, 180
\GTSnDADM	17, 372, 379, 381, 382, 383
\GTSnDADMR	18, 378
\GTSnDADR	10, 176
\GTSnDAU	9, 159, 163, 165, 166
\GTSnDAUM	16, 348, 355, 357, 358, 359
\GTSnDAUMR	16, 354
\GTSnDAUR	9, 162
\GTSnegatedDirectedAdjacencyDown	10, 173
\GTSnegatedDirectedAdjacencyDownMonochrome	17, 372
\GTSnegatedDirectedAdjacencyDownMonochromeRelation	18, 378
\GTSnegatedDirectedAdjacencyDownRelation	10, 176
\GTSnegatedDirectedAdjacencyUp	9, 159
\GTSnegatedDirectedAdjacencyUpMonochrome	16, 348
\GTSnegatedDirectedAdjacencyUpMonochromeRelation	16, 354
\GTSnegatedDirectedAdjacencyUpRelation	9, 162
\GTSnegatedIncidence	12, 290
\GTSnegatedIncidenceMonochrome	18, 392
\GTSnegatedIncidenceMonochromeRelation	19, 398
\GTSnegatedIncidenceRelation	12, 293
\GTSnegatedInIncidence	14, 314
\GTSnegatedInIncidenceMonochrome	21, 430
\GTSnegatedInIncidenceMonochromeRelation	21, 436
\GTSnegatedInIncidenceRelation	14, 317
\GTSnegatedOutIncidence	13, 302
\GTSnegatedOutIncidenceMonochrome	20, 410
\GTSnegatedOutIncidenceMonochromeRelation	20, 416
\GTSnegatedOutIncidenceRelation	13, 305
\GTSnegDirAdjDown	10, 173
\GTSnegDirAdjDownMon	17, 372
\GTSnegDirAdjDownMono	17, 372
\GTSnegDirAdjDownMonoRel	18, 378
\GTSnegDirAdjDownMonRel	18, 378
\GTSnegDirAdjDownRel	10, 176
\GTSnegDirAdjUp	9, 159
\GTSnegDirAdjUpMon	16, 348
\GTSnegDirAdjUpMono	16, 348
\GTSnegDirAdjUpMonoRel	16, 354

\GTSnegDirAdjUpMonRel	16, 354
\GTSnegDirAdjUpRel	9, 162
\GTSnegInc	12, 290
\GTSnegIncMon	18, 392
\GTSnegIncMono	18, 392
\GTSnegIncMonoRel	19, 398
\GTSnegIncMonRel	19, 398
\GTSnegIncRel	12, 293
\GTSnegInInc	14, 314
\GTSnegInIncMon	21, 430
\GTSnegInIncMono	21, 430
\GTSnegInIncMonoRel	21, 436
\GTSnegInIncMonRel	21, 436
\GTSnegInIncRel	14, 317
\GTSnegOutInc	13, 302
\GTSnegOutIncMon	20, 410
\GTSnegOutIncMono	20, 410
\GTSnegOutIncMonoRel	20, 416
\GTSnegOutIncMonRel	20, 416
\GTSnegOutIncRel	13, 305
\GTSnI	12, 290 , 293 , 294 , 295
\GTSnII	14, 314 , 317 , 318 , 319
\GTSnIIM	21, 430 , 437 , 439 , 440 , 441
\GTSnIIMR	21, 436
\GTSnIIR	14, 317
\GTSnIM	18, 392 , 398 , 399 , 400 , 401
\GTSnIMR	19, 398
\GTSnIR	12, 293
\GTSnOI	13, 302 , 305 , 306 , 307
\GTSnOIM	20, 410 , 417 , 419 , 420 , 421
\GTSnOIMR	20, 416
\GTSnOIR	13, 305
\GTSoI	12, 296 , 299 , 300 , 301
\GTSoIM	19, 402 , 406 , 407 , 408 , 409
\GTSoIMR	19, 406
\GTSoIR	12, 299
\GTSoutInc	12, 296
\GTSoutIncidence	12, 296
\GTSoutIncidenceMonochrome	19, 402
\GTSoutIncidenceMonochromeRelation	19, 406
\GTSoutIncidenceRelation	12, 299
\GTSoutIncMon	19, 402
\GTSoutIncMono	19, 402
\GTSoutIncMonoRel	19, 406
\GTSoutIncMonRel	19, 406
\GTSoutIncRel	12, 299
\GTSsetMainColor	7, 24 , 36 , 86 , 183 , 229
\GTSsetSecondColor	7, 29 , 39 , 89 , 186 , 232
\GTSu	8, 147 , 150 , 151 , 152
\GTSuM	15, 328 , 332 , 333 , 334 , 335
\GTSuMR	15, 332
\GTSunadj	8, 147
\GTSunadjacency	8, 147
\GTSunadjacencyMonochrome	15, 328
\GTSunadjacencyMonochromeRelation	15, 332
\GTSunadjacencyRelation	8, 150
\GTSunadjMon	15, 328
\GTSunadjMono	15, 328
\GTSunadjMonoRel	15, 332
\GTSunadjMonRel	15, 332
\GTSunadjRel	8, 150

\GTSuR	8, <u>150</u>
\GTSx	23, <u>517</u>
\GTSxa	7, <u>99</u> , 519
\GTSxadj	7, <u>99</u>
\GTSxadjacency	7, <u>99</u>
\GTSxi	11, <u>242</u> , 521
\GTSxinc	11, <u>242</u>
\GTSxincidence	11, <u>242</u>
\GTSxmetallogic	23, <u>479</u>
\GTSxml	23, <u>479</u> , 523